Agricultural Water Management 98 (2010) 331–342

Contents lists available at ScienceDirect

Agricultural Water Management

journal homepage: www.elsevier.com/locate/agwat

Hydrological impacts of rainwater harvesting (RWH) in a case study catchment: The Arvari River, Rajasthan, India. Part 1: Field-scale impacts

C.J. Glendenning [a,*], R.W. Vervoort [b,1]

[a] International Food Policy Research Institute, NASC Complex, CG Block, Pusa, New Delhi 110 012, India
[b] Hydrology Research Laboratory, Faculty of Agriculture, Food and Natural Resources The University of Sydney, NSW 2006, Australia

ARTICLE INFO

Article history:
Received 15 March 2010
Accepted 2 September 2010
Available online 26 September 2010

Keywords:
Water storage
Groundwater
Recharge
Water balance
Water table fluctuation

ABSTRACT

Rainwater harvesting (RWH), the small-scale collection and storage of runoff to augment groundwater stores, has been seen as a solution to the deepening groundwater crisis in India. However, hydrological impacts of RWH in India are not well understood, particularly at the larger catchment-scale. A key element to grasping RWH impact involves understanding the generated recharge variability in time and space, which is the result of variability in rainfall-runoff and efficiency of RWH structures. Yet there are very few reported empirical studies of the impact of RWH. Catchment-scale impacts are best studied using a water balance model, which would require a basic level of field data and understanding of the variability. This study reports the results of a 2-year field study in the 476 km^2 semi-arid Arvari River catchment, where over 366 RWH structures have been built since 1985. Difficulties associated with working in semi-arid regions include data scarcity. Potential recharge estimates from seven RWH storages, across three different types and in six landscape positions, were calculated using the water balance method. These estimates were compared with recharge estimates from monitored water levels in 29 dug wells using the water table fluctuation method. The average daily potential recharge from RWH structures varied between 12 and 52 mm/day, while estimated actual recharge reaching the groundwater ranged from 3 to 7 mm/day. The large difference between recharge estimates could be explained through soil storage, local groundwater mounding beneath structures and a large lateral transmissivity in the aquifer. Overall, approximately 7% of rainfall is recharged by RWH in the catchment, which was similar in the comparatively wet and dry years of the field analysis. There were key differences between RWH structures, due to engineering design and location. These results indicate that recharge from RWH affects the local groundwater table, but also has potential to move laterally and impact surrounding areas. However, the greatest weakness in such analysis is the lack of information available on aquifer characteristics, in addition to geology and soil type.

© 2010 Elsevier B.V. All rights reserved.

1. Introduction

Groundwater is an important resource for humans. Globally groundwater provides 50% of drinking water and 20% of irrigation water (Villholth, 2006). In India, where 15% of the global population lives, 55–60% of the Indian population is directly or indirectly dependent on groundwater for its livelihood (Villholth, 2006). Since independence in 1947, the technological advance in groundwater pumping from draft power and human labour to diesel pumps and electricity has greatly impacted the amount of extraction (Shah et al., 2006). As a direct result of the increased use of groundwater in India, millions of people have been lifted out of poverty (Kemper, 2007). However, while exponential groundwater use over the past few decades has improved stability in cropping (Shah, 2007), there are increasingly serious issues with aquifer depletion, as groundwater tables are rapidly falling in a number of states (Rodell et al., 2009; Shah et al., 2007). Given that, in addition to increasing population pressures, the severity of extreme events such as droughts and floods is predicted to increase over the next 20 years, groundwater management will need to be a major focus in India (Gupta and Deshpande, 2004; Pandey et al., 2005; Ramesh and Yadava, 2005). Research into feasible solutions to sustain groundwater stores is therefore gaining considerable momentum, and rainwater harvesting (RWH) is encouraged as a possible solution for sustainable groundwater management.

RWH is a traditional catchment development tool in South Asia. It involves a small dam that collects and stores runoff during the heavy downpours of the Indian monsoon season, allowing time for the stored water to recharge shallow groundwater aquifers. Rainwater harvesting is a managed aquifer recharge technique, which aims to increase groundwater supply. Because RWH structures tend

* Corresponding author. Tel.: +91 011 2584 6565; fax: +91 011 2584 8008.
E-mail address: c.glendenning@cgiar.org (C.J. Glendenning).
[1] Tel.: +61 02 8627 1054.

0378-3774/$ – see front matter © 2010 Elsevier B.V. All rights reserved.
doi:10.1016/j.agwat.2010.09.003

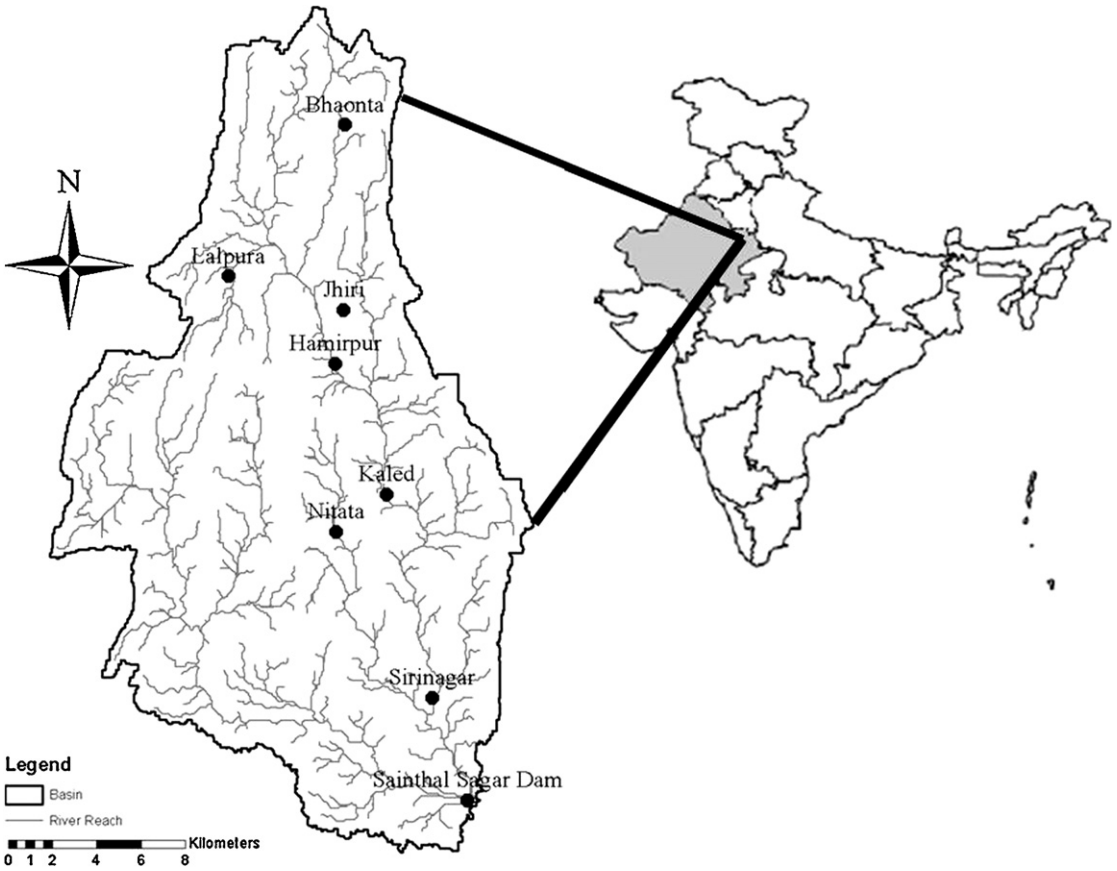


Fig. 1. Position of the Arvari River catchment in the eastern part of the state of Rajasthan, India. Villages, which were the focus of the data collection, are highlighted.

to have a small storage capacity, with an average capacity between 0.4 and 11×10^{-4} m^3 (Sharda et al., 2006), they respond faster to rainfall-runoff than larger dams. Rainwater harvesting generally results in increased crop production and greater crop yield, because resulting rises in the water table mean better accessibility and yields of groundwater (Keller et al., 2000). The 'groundwater recharge movement' in India has been promoting and reinvesting in RWH over the past three decades (Sakthivadivel, 2007, 2008).

Until recently, RWH was seen as a totally benign technology, as the impact of an individual dam was considered relatively small (Batchelor et al., 2003). However, there is a lack of systematic studies on the hydrological impact of RWH structures and associated recharge on catchment-scale water balances. This despite the fact that the cumulative impact of RWH structures on stream flows could be significant (Calder et al., 2006). The need to quantify the overall impact of RWH in a catchment is important, because it has the potential to cause unplanned impacts, such as inequitable sharing of water between upstream and downstream users (Batchelor et al., 2002; Kumar et al., 2006). Studies have highlighted the potential negative impacts of RWH, including low reliability in drought years and smaller benefits to augment groundwater stores at higher degrees of catchment development (Kumar et al., 2008). Rainwater harvesting redistributes the available water resources across the catchment, by increasing the annual amount of catchment rainfall that becomes stored recharge, which changes other water balance components, including evaporation and stream flow. The arguments against RWH have not yet been confirmed by any empirical studies. This is because the few empirical studies in India have focussed largely on local-scale RWH impacts and have not considered larger catchment hydrological impacts, or the surface water–groundwater interactions (Badiger et al., 2002; Gontia and Sikarwar, 2005; Neumann et al., 2004; Sharda et al., 2006). Consequently, despite the potential positive impact of RWH for irrigated agriculture through increased groundwater availability, there is a serious lack of understanding of catchment-scale impacts of RWH (Barah, 1996; Kumar et al., 2006, 2008; Sakthivadivel, 2007; Shah, 2007).

In summary, there are only limited studies and estimates of local scale recharge from RWH structures (e.g. Badiger et al., 2002; Sharda et al., 2006) and the resulting catchment-scale impacts. The overall study reported here therefore sets out to achieve two aims. The first is to better understand the impact of RWH on local water balances and the factors affecting RWH recharge through field observations. The second was to extrapolate these field observations to the catchment-scale using a conceptual water balance model. This first paper reports on the field studies.

2. Study area

2.1. Location of the Arvari River catchment

The case study area is the Arvari River catchment, which is located in the state of Rajasthan (Fig. 1). Rajasthan is the largest state in India and is predominantly agrarian: 70% of its population depends on agriculture-based activities. Although the state covers 10.5% of India's geographical area, it shares only 1.2% of its water resources (Narain et al., 2005).

The Arvari River catchment (476 km^2) is in the eastern part of the state (latitude 27°1′30″ N and 27°22′15″ N, longitude 76°16′30″ E and 76°4′55″ E), in the district of Alwar, and is a tributary covering 7% of the larger Banganga River Basin. The catchment drains into Sainthal Sagar dam, a medium size irrigation project built in 1898. While the inflows into the dam come from two semi-arid ephemeral rivers, the Bidila and the Arvari, the catchment is com-

monly referred to as the Arvari River. Geologically it is a part of the Aravalli Hill range, which is one of the oldest mountain ranges in the world (Bhuiyan et al., 2006).

In the catchment there has been significant reinvestment in RWH over the last 20 years, which is the result of the work of the community and a local non-government organisation (NGO), Tarun Bharat Sangh. As a result, this makes it an interesting case study to evaluate the impacts of RWH on a larger scale. However, as is typical for many semi-arid catchments, the catchment is ungauged and there is no climate station within the catchment. Additionally existing information on aquifer properties, geology and soils is at a very large scale, which does not capture the spatial and temporal variability at finer scales within the catchment.

2.2. Climate

The climate is semi-arid with hot summers between March and June. May and June are the hottest months of the year with a mean daily maximum and minimum temperature in May of 42.1 and 25.7 °C. The winter extends from November to March, with temperatures dropping to an average minimum of 7.7 °C and an average maximum of 21 °C in January. Eighty percent of the rainfall falls during the south-westerly monsoon between June and September. The closest rainfall station to the Arvari River catchment is in the nearby town of Thanagazi (15 km to the North of the catchment), which has an average yearly rainfall of 705.8 mm. Each year there is an average of 8 rainfall days greater than 35 mm with an average of 31 rainy days a year (1901–2002). Total annual potential evapotranspiration is 1523.7 mm, highest in May (271.5 mm/month) and lowest in December (155.6 mm/month).

2.3. Geology, hydrogeology and soils

In the Arvari River catchment, the geology consists of high relief basement rocks up to 750 m above sea level (ASL) and with sediment of varying thickness in the valleys (300–400 m ASL) (COMMAN, 2005a). In the greater Banganga River Basin, groundwater occurs in shallow alluvium layers and deeper hard rock areas. The sandy and coarse sediments of the alluvial system overly the fractured igneous and metamorphic hard rock aquifers, and is connected hydraulically (Gale et al., 2002).

Examined soil profiles in the catchment were dominated by a shallow, well-drained, loamy skeletal soil, which is severely eroded and strongly stony. Saprolite is commonly seen throughout the catchment and bedrock is reached at shallow depths. Land use reflects soil type, with agriculture found on the alluvial deeper clay soils, while forest areas were on the slopes on shallower soils. There is limited small-scale information about geological and soil characteristics in this catchment.

2.4. Rainwater harvesting (RWH) structures

The community, with the local NGO, Tarun Bharat Sangh, has built around 366 RWH structures in the Arvari River catchment since 1985. The structures are found throughout the catchment (Fig. 2). The different structure types, shown in Fig. 3, have different physical specifications (Table 1) and include:

- *ANICUTS* are built on common land and dam the main reach of the river. They are generally made of cement and stone or concrete. These structures have a very large impact on local groundwater tables (Fig. 3a).
- *BANDHS* or small dams are similar to *Anicuts*. *Bandhs* dam tributaries on the main river reach and are made of concrete in the middle with earthen outer edges, while some are entirely made of earth (Fig. 3b).
- *JOHADs* and *JOHADIs* are small earthen dams shaped like a crescent moon. They are found at the foothills of slopes, collecting water from a small hilly catchment area. The main purposes of *Johads* are for livestock drinking and bathing. *Johadis* are smaller *Johads* (Fig. 3c).
- *TALABs* and *TALAIs* are deep holes or pond-like structures that have high raised edges on 3 sides made of earth, and are similar to *Johads*. Monsoon water is collected in the *Talab* for village use and livestock drinking. *Talais* are smaller *Talabs*.

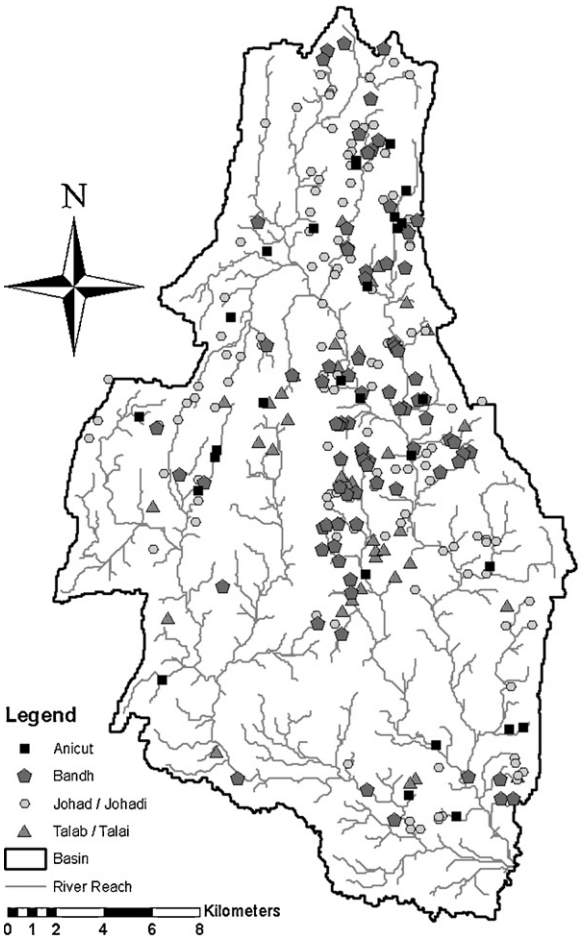


Fig. 2. The different types of RWH structures and their extent within the Arvari River catchment.

Low in-field walls or *MEDBANDHIs*, mainly aimed at increasing in-field soil moisture, were not included in the analysis as their contribution to recharge was considered minimal.

Table 1
Average physical specifications of RWH structures built by Tarun Bharat Sangh (TBS) and the community since 1985–2005 in the Arvari River catchment (pers. comm. TBS).

Structure type	Number	Catchment area (ha)	Storage surface area (ha)	Maximum storage depth (m)
Anicut	28	1090	3	2.7
Bandh	84	460	4	3.5
Johad/Johadi	127	54	1.5	3
Talab/Talai	39	30	1	3

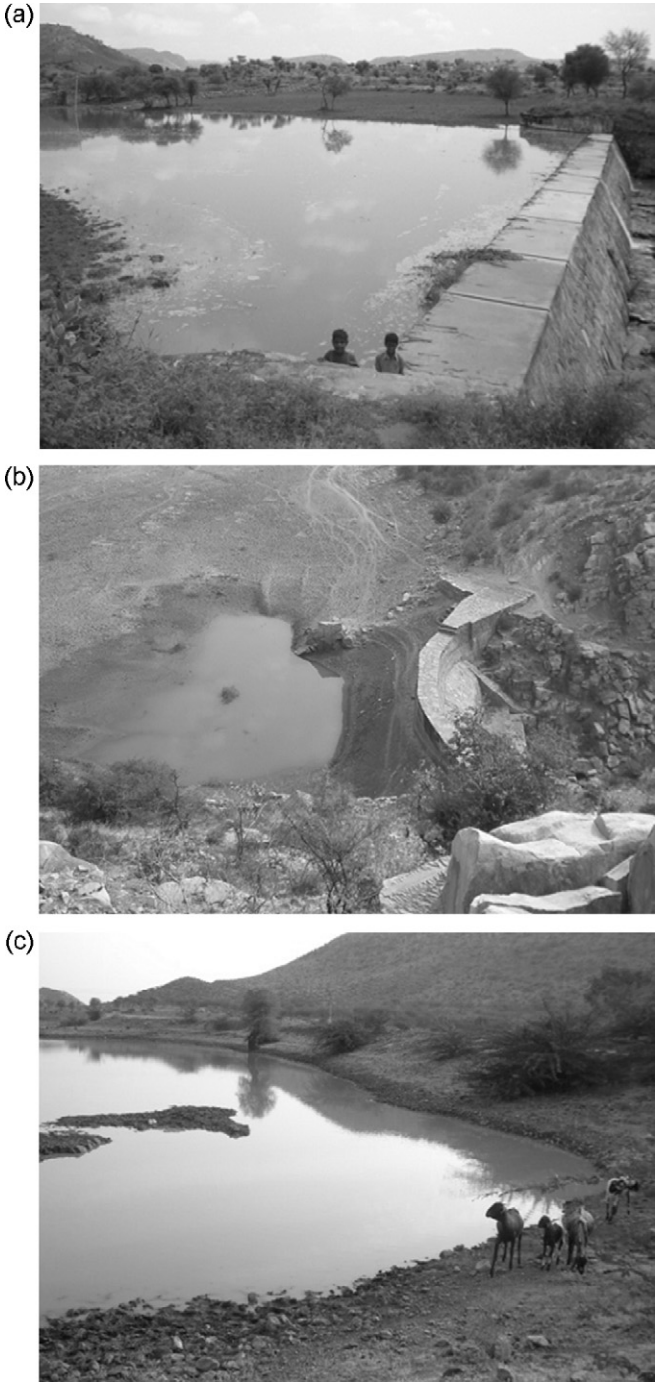


Fig. 3. Example rainwater harvesting (RWH) structures present in the Arvari River Catchment: (a) *Anicut*; (b) *Bandh*; (c) *Johad*.

3. Methods

3.1. Data collection network

Six villages in the middle and upper reaches of the Arvari catchment were identified for detailed studies of wells and RWH structures over 2007 and 2008, based on position and accessibility. These focus areas are located mainly in the middle and upper reaches of the catchment (Fig. 1, Table 2). In each focus area a RWH structure was monitored, along with dug wells in the vicinity of the structure (Table 2). In Bhaonta, two cascading *Bandhs* were monitored, in Lalpura and Jhiri earthen *Johads* were monitored, and finally in Hamirpur, Nitata and Kaled *Anicuts* that lie across the main reach of the river were observed. As there are no official instrumented piezometers in the catchment, groundwater levels in 29 dug wells were monitored throughout 2007 and 2008. Three rain gauges were set up in Bhaonta, Hamirpur and Sirinagar to collect daily rainfall (Fig. 1).

3.2. Potential recharge from RWH structures

On non-rainy days, potential recharge (R_{ep} m^3) from RWH structures can be calculated using the water balance approach assuming runoff into the structures is negligible (Sharda et al., 2006):

$$R_{ep} = A_s(\Delta h - \mathrm{ET}) - O_t \quad (1)$$

where Δh = change in depth (m) in water level of the structure over 2 days ($h_t - h_{t-1}$), A_s = estimated average surface area (m^2) of stored water over 2 days, using the Heronian mean, i.e. assuming a cone shaped water body: $(A_{st} + A_{st-1} + (A_{st} - A_{st-1})^{1/2})/3$, ET = evaporation (m), which uses the monthly average daily potential evaporation rate for the Alwar district (1901–2002) (India Water Portal, 2009), which means $A_s(\Delta h - ET)$ represents the change in storage. Finally O_t = overflow (m^3).

To calculate potential R_{ep} on rainy days, an empirical relationship between the average depth of water stored in the structures and R_{ep} on non-rainy days, calculated from Eq. (1), (Sharda et al., 2006) was fitted:

$$R_{ep} = a h_{av}^{b} \quad (2)$$

where a and b are curve fitting parameters and h_{av} = average depth on non-rainy days ($h_t + h_{t-1}/2$). This relationship was subsequently used to estimate the volume of R_{ep} on rainy days using the measured depth on those days.

3.3. Recharge efficiency index of RWH structures

The annual hydrogeological efficiency or recharge efficiency ($\eta_{recharge}$) of the RWH structures, can be based on the relationship between the total annual volume of R_{ep} (m^3) and the total annual volume of water stored in the structure ($V_{run\text{-}on}$ m^3) (Beernaerts, 2006; Martin-Rosales et al., 2007):

$$\eta_{recharge} = \frac{R_{ep}}{V_{run-on}} \quad (3)$$

3.4. Recharge based on groundwater levels in dug wells

The groundwater levels in the monitored dug wells can be used to estimate recharge using the water table fluctuation (WTF) method. Assuming a constant specific yield (S_y), recharge (R_{gw} mm/day) can be estimated from the rise in the water table (Δh) over time (Δt) for each monitored dug well (Healy and Cook, 2002; Scanlon et al., 2002):

$$R_{gw} = \frac{S_y \, \Delta h}{\Delta t} \quad (4)$$

Based on literature a constant specific yield (S_y) of 0.1 was chosen (COMMAN, 2005b), assuming an unconsolidated geological formation. The lithology of the area is predominantly alluvium and blown sand, and standard published S_y values indicate similar estimates (Johnson, 1967). Assuming a constant S_y can introduce errors in recharge estimations (Sophocleous, 1991), but there is little other data to allow a more accurate estimation of S_y, such as used in other studies (Crosbie et al., 2005; Marechal et al., 2006; Muralidharan and Andrade, 2007; Wendland et al., 2007).

Table 2
Name and specifications of monitored RWH structures for each focus area of the field study in the Arvari River catchment in 2007 and 2008.

Village	Structure name	RWH type	Maximum storage area (ha)	Maximum storage depth (m)	Capacity (10^{-4} m^3)	Catchment area (ha)
Bhaonta	Sankara	*Bandh*	2.9	8	23.4	200
	Beruji	*Bandh*	1.9	4	7.8	300
Lalpura	Koda Wala	*Johad*	2.0	3	7.1	150
Jhiri	Amli	*Johad*	1.4	3	4.2	Not known
Hamirpur	Jabar Sagar	*Anicut*	1.8	3	5.0	10 500
Kaled	Tode Sagar	*Anicut*	2.3	2	4.6	15 200
Nitata	Bhoja Kala	*Anicut*	0.7	1.5	1.1	300

Table 3
Annual rainfall (mm) and number of rainy days in the Arvari River catchment.

Focus area	Total rainfall (mm) 2007	Total rainfall (mm) 2008	Rainy days 2007	Rainy days 2008
Bhaonta	361	751	40	32
Hamirpur	449	897	27	56
Sirinagar	499	494	18	38
Average	436	714	28	42

4. Results

4.1. Catchment rainfall variation

Rainfall in the catchment is characterised by high variability and highly localised events. In 2007, the lower end of the catchment at Sirinagar received the greatest amount of rainfall, with one large event of over 100 mm. In contrast, in 2008, the upper catchment received much more rain and had more rainy days with greater event sizes than the lower catchment at Sirinagar (Table 3). The magnitude of the daily rainfall was also very different between gauges (Fig. 4). In 2008 there were a greater number of days with rainfalls greater than 10 mm.

The data from the catchment rain gauges thus displays high spatial variability during the monsoon season in both event size and temporal distribution (Table 3, Fig. 4). Compared to the longer term rainfall data from nearby Thanagazi (1902–2002), the 2007 average annual catchment rainfall was below average with a smaller number of large events. In contrast, 2008 had higher average annual rainfall; the monsoon season was spread over a longer period of time and had larger daily rainfall amounts. The rainfall characteristics will have an impact on the functioning of RWH structures as storages require larger rainfall events to fill (Muralidharan and Andrade, 2007). Nevertheless, it should be noted that rainfall is highly localised, so that some structures may receive more rainfall than others within an area.

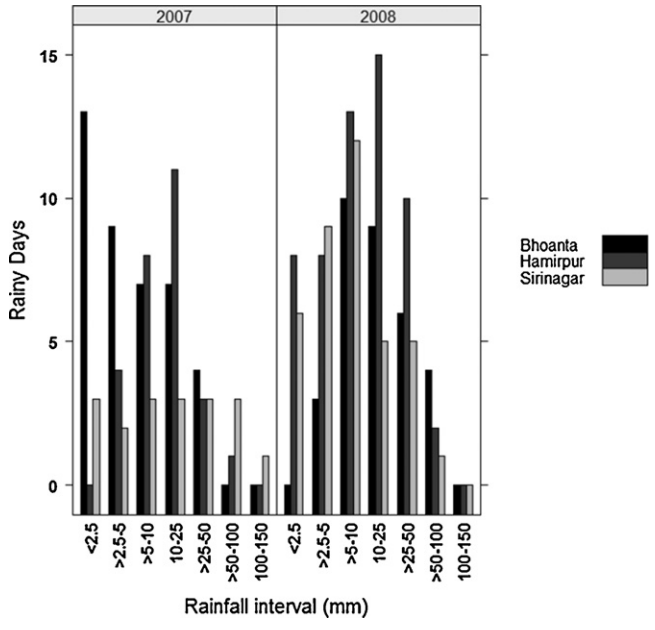


Fig. 4. Variation in daily rainfall depth intervals (mm) in the Arvari River catchment during the monsoon season (June–September) of 2007 and 2008.

4.2. Potential recharge volume (R_{ep}) estimates from RWH structures using the water balance method

Statistical parameters for the fits of Eq. (2) for each structure are shown in Table 4. As pointed out in Section 3, the R_{ep} calculated on non-rainy days was used to estimate the volume of R_{ep} on rainy days based on the relationship of R_{ep} with average depth (Sharda et al., 2006). The noise in the data, as indicated by the root mean square error (RMSE) and the standard deviation (Table 4), is a reflection of the difficulties associated with estimating the planar surface area of the structures, which had highly irregular shapes.

Total volume of R_{ep} was greatest at Sankara *Bandh* (Fig. 5a). Beruji *Bandh* was directly below Sankara *Bandh*, so the volume of recharge here is less, as less runoff reached this structure and average storage depth was lower (Fig. 5b). *Bandh*s also had a shorter storage time than the *Johad*s (Table 5). Beruji *Bandh* emptied the fastest of all structures, probably mainly due to the smaller run-on volume, but also possibly due to the underlying geology, which appeared to have higher infiltration rates. The monsoon season extended longer and rainfall was greater in 2008, which is reflected in the R_{ep} values with a longer time series and greater volume compared to 2007 (Fig. 5).

Potential recharge was converted from m^3/day to mm/day by dividing by the average surface area of the structures. The average daily R_{ep} rates (mm/day) (Table 5) in 2008 are much greater than in 2007, as more rainfall in 2008 meant more stored water in the RWH storage areas. The rate of potential recharge depends on the shape and size of the structure storage area, runoff characteristics (local rainfall and land cover), and infiltration characteristics in the storage area (soil depth and underlying bed rock). In the *Johad*s, the soil depth was deeper and had higher clay content compared to the *Bandh*s, which would influence the infiltration capacity of the stored water. For all structures the total volume of R_{ep} increases as the rainfall increases. As only 2 years of field data has been collected, it is difficult to determine the exact relationship between annual rainfall and annual recharge. However, the observed data indicate the range of R_{ep} values that can occur across a variable landscape. The type of structure determines the range of potential recharge values. This can probably be explained by the fact that the structures types were purpose built to fulfil different objectives, which results in similar properties across the catchment.

Studies conducted on recharge through percolation tanks constructed in hard rock and alluvium formations in the Pali district (Western Rajasthan), showed percolation rates of 14–52 mm/day (Narain et al., 2005), which are similar to this study. A theoretical study by Neumann et al. (2004) compared different evaporation rates with different recharge rates from structures. Percolation rates similar to evaporation rates (3.6 mm/day) suggest that the

Table 4
Statistical parameters for Eq. (2), describing the relationship between average depth A_s (m) in RWH structures and potential recharge volume R_{ep} (m^3) on non-rainy days. These parameters are used to calculate potential recharge volume R_{ep} (m^3) on rainy days.

Village	RWH structure	*a*	*b*	R^2	RMSE (m^3)	Standard deviation of R_{ep} (m^3)
Bhaonta	Sankara *Bandh*	179.9	1.7	0.93	1601.3	873.1
	Beruji *Bandh*	69.3	2.2	0.69	427.2	115.2
Jhiri	Jhiri *Johad*	82.1	1.8	0.62	536.8	403.9
Lalpura	Lalpura *Johad*	117.2	3.5	0.95	818.8	658.0

Table 5
Average daily recharge R_{ep} (mm/day) and number of days water was stored in monitored RWH structures in 2007 and 2008.

RWH Structure	Year	Total days of storage	Average daily R_{ep} (mm/day)	Standard deviation of R_{ep} (mm/day)
Sankara *Bandh*	2007	169	45.6	0.07
	2008	206	55.6	0.15
Beruji *Bandh*	2007	138	20.9	0.07
	2008	201	27.8	0.03
Jhiri *Johad*	2007	273	12.3	0.01
	2008	308	19.5	0.01
Lalpura *Johad*	2007	240	15.7	0.02
	2008	382	23.5	0.04

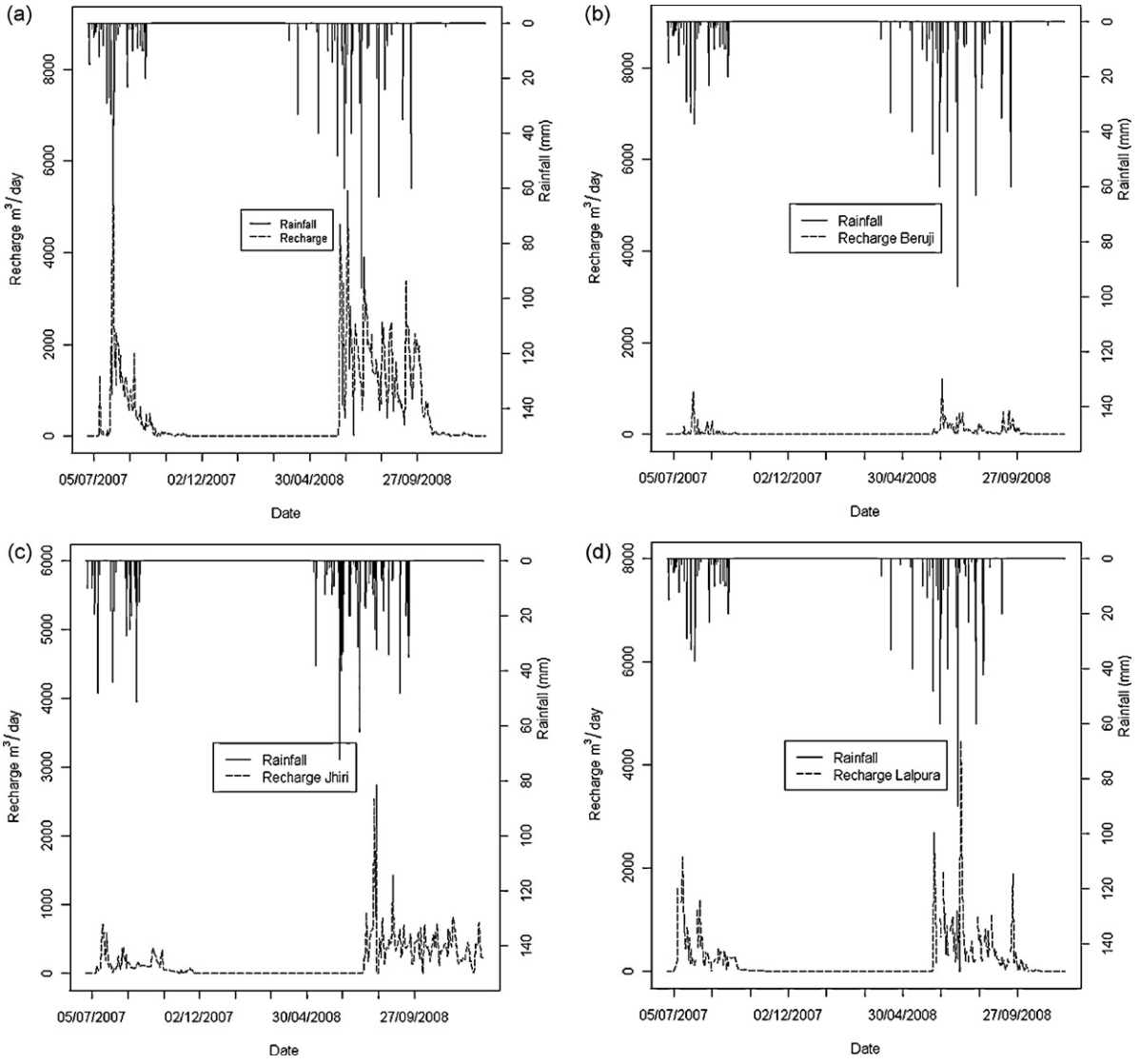


Fig. 5. Estimated volume of potential recharge (R_{ep} m^3/day) with daily rainfall (mm) from Bhaonta gauge for (a) Sankara *Bandh*, (b) Beruji *Bandh*, and with daily rainfall (mm) from Hamirpur gauge for (c) Jhiri *Johad*, (d) Lalpura *Johad*.

Table 6
Statistical parameters for Eq. (5) describing the relationship between the cumulative sum of recharge R_{ep} (mm) and cumulative sum of rainfall (mm).

Structure	R^2	a	b	c
Sankara *Bandh*	0.89	2.78	1.94	0.004
Beruji *Bandh*	0.88	2.86	0.21	0.005
Jhiri *Johad*	0.93	2.78	0.75	0.003
Lalpura *Johad*	0.81	2.68	0.77	0.008

structures would be inefficient (Neumann et al., 2004), while structures with higher water level decline rates of 24 mm/day or greater, as found in this study, suggest more recharge (Table 5).

4.3. Potential recharge and rainfall relationship

Despite the highly local characteristics of rainfall, a relationship between R_{ep} from RWH structures and rainfall using the closest rainfall gauge to each structure, can determine how much rainfall would be needed for a given amount of R_{ep}. As suggested by Sharda et al. (2006), a Rosenbrock–Quasi Newton function was fitted to the 2-year time series to describe the relationship between cumulative rainfall and cumulative R_{ep}:

$$\log_{10}(R_{ep}) = a[b - e^{-cP}] \quad (5)$$

where R_{ep} = cumulative recharge (mm), P = cumulative rainfall (mm), a, b and c = parameters. The parameters and R^2 describing the fits of Eq. (5) are listed in Table 6.

In all structures R_{ep} reaches a maximum daily depth limit as cumulative rainfall continues to infinity, a relationship also described in Sharda et al. (2006). This reflects the engineering design of the structures, which can only store a certain amount of runoff, and so can only induce a maximum daily depth of R_{ep}.

Using the regression fits in Table 6 and Eq. (5), the cumulative rainfall required to induce 1, 10 and 100 mm of R_{ep} from the monitored RWH structures can be calculated and compared with the results in Sharda et al. (2006). In Fig. 6, Beruji *Bandh*, Jhiri *Johad* and Lalpura *Johad* display similar results to the estimates of Sharda et al. (2006). However, Sankara *Bandh* requires less cumulative rainfall to induce recharge. Sankara *Bandh* has a larger storage, and

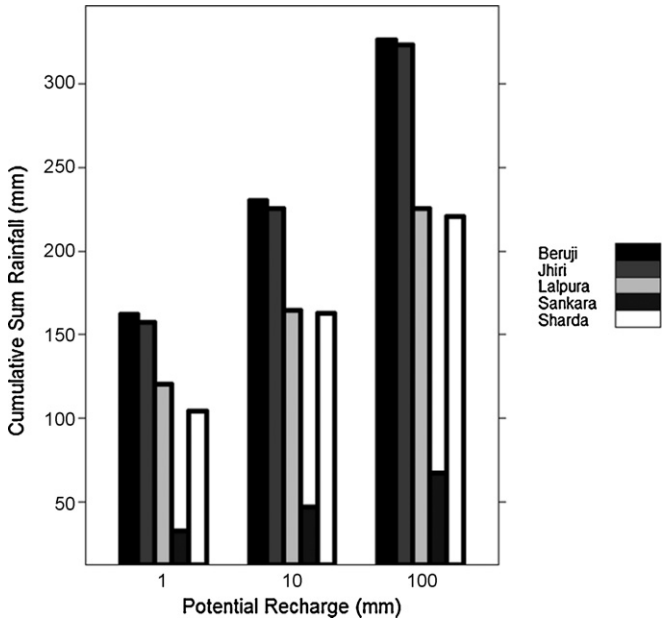


Fig. 6. Cumulative rainfall (mm) required for 1, 10 and 100 mm of potential recharge (R_{ep} mm/day) for each monitored RWH structure and compared with estimates from Sharda et al. (2006).

Table 7
Recharge efficiency of RWH structures in 2007 and 2008.

RWH structure	$\eta_{recharge}$ % 2007	$\eta_{recharge}$ % 2008
Sankara *Bandh*	6.0	8.2
Beruji *Bandh*	16.4	14.1
Jhiri *Johad*	2.1	1.3
Lalpura *Johad*	3.4	4.1

is situated in the steep rocky hills, where the percolation rate is highest (Table 5). Beruji *Bandh* requires more cumulative rainfall to induce recharge, which is a reflection of the position of Beruji *Bandh*, below Sankara *Bandh* (which would first collect runoff). The *Johad*s, Lalpura and Jhiri, hold water in storage for the longest time due to lower infiltration rates (Table 5). As a result, more rainfall is needed for these structures to induce similar R_{ep} to the *Bandhs*. These results highlight the difference in R_{ep} between the larger and smaller RWH structures, and the importance of underlying geology and soil depths in RWH functioning.

4.4. Recharge efficiency of RWH structures

There is a not a large difference in recharge efficiencies between the monitored years (Table 7). In 2007 the volume stored in the structures was much less than in 2008, but this does not influence recharge efficiency, as the ratio of $V_{run\text{-}on}$ to volume of R_{ep} is similar across the 2 years. The most efficient RWH structure is Beruji *Bandh*, probably because it had the shortest storage time due to high infiltration rates and therefore the least evaporative losses (Table 5). In 2008 there is a greater storage surface area in all the RWH structures, but at the exteriors of the storage area are shallower depths. The larger surface area means greater evaporation losses, as the high surface area to volume ratio of small reservoirs leads to higher evaporation loss (Keller et al., 2000). This is possibly why Sankara *Bandh*, with the highest potential recharge volume, but also the largest storage surface area (and therefore higher evaporative losses), has lower recharge efficiency than Beruji *Bandh*. In contrast, the monitored *Johad*s have similar, but lower, recharge efficiency. The *Johad*s at Jhiri and Lalpura held water in storage longer than in the *Bandhs*, and so evaporative losses would have been higher, reducing the recharge efficiency (Table 5).

4.5. River flow and storage

As discussed in Section 3, discharge across three *Anicut*s that lie across the main reach of the river was also monitored. In 2007, the *Anicut* at Hamirpur overflowed on 8 days towards Kaled. At Kaled *Anicut*, overflow occurred for 22 days and further downstream towards Sainthal Sagar dam, at Sirinagar, there was no stream flow (see Fig. 1 for catchment position). On the other tributary, at Nitata, the *Anicut* overflowed on 6 days. In 2008, the increased rainfall led to more flow; Hamirpur 24 days, Nitata 11 days, but Kaled had the longest overflow, flowing daily through the months of July to early November. The peaks occurred in all the *Anicut*s at similar times, mostly when large rainfall events occurred and therefore greater runoff events reached the river.

Given that no surface areas were measured for the water stored in the *Anicut*s, only the R_{ep} rate (mm/day) from the *Anicut*s was estimated using the water balance method for days when there was no overflow, no rainfall and no runoff in the storage:

$$R_{ep} = \Delta h - ET \quad (6)$$

where R_{ep} = potential recharge (mm/day), Δh = change in depth behind dam wall (mm/day), and ET = potential evaporation (mm/day).

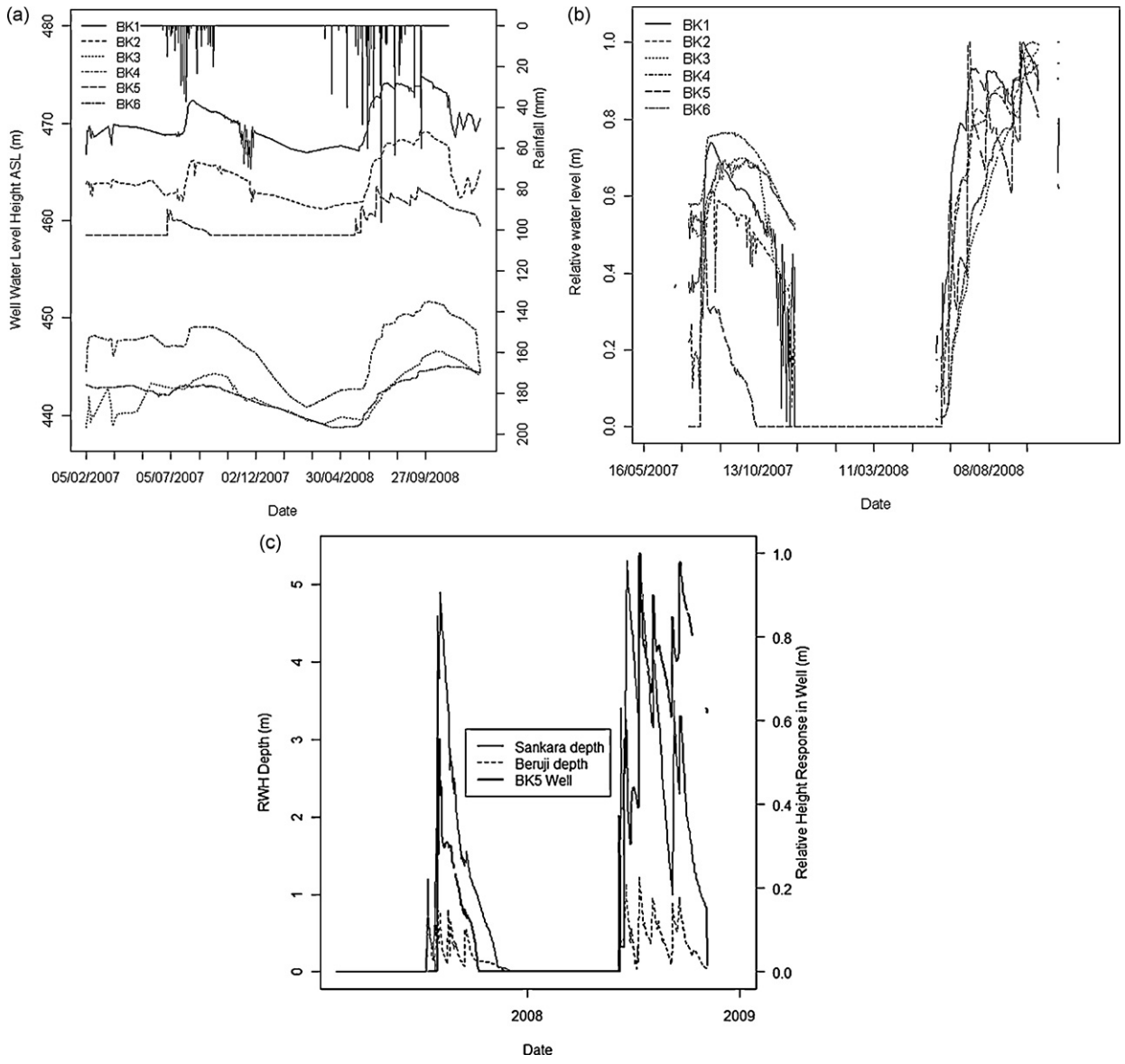


Fig. 7. The six monitored wells at focus area Bhaonta: (a) ASL (m) water level; (b) relative water level height; (c) relative well height at well BK5 and depths at Sankara *Bandh* and Beruji *Bandh* (m).

Average daily recharge varied from 42 mm/day at Nitata, to 25 mm/day at Kaled and 44 mm/day at Hamirpur. This analysis assumes that R_{ep} occurs evenly over the storage surface area. As a result, the volumetric total R_{ep} would vary as this would be related to the surface area. In addition R_{ep} could only be estimated once overflow had stopped, so actual R_{ep} could be much higher, which explains the lower values at Kaled *Anicut*. Nevertheless the estimated R_{ep} values are high compared to the values obtained from the *Johads* and *Bandhs*, possibly due to differences in the underlying stratum. This confirms the community's observation that the structures on the river are important for recharge. These results could also provide approximate transmissions losses from the river reaches.

4.6. Well response

Most wells had diesel pumps and these were used extensively for irrigation during the monsoon of 2007 due to the low rainfall. Consequently water levels in pumped wells decreased significantly after reaching a maximum height early in the monsoon of 2007. This made it difficult to estimate recharge using the water table fluctuation (WTF) method. Rainfall in 2008 was mostly sufficient for meeting crop needs, resulting in less pumping, so the water level rises in the wells can be seen more clearly in 2008. In Figs. 7–9, the results from 17 of the 29 monitored dug wells are highlighted. Groundwater levels are plotted as heights above sea level (ASL) and also as relative groundwater levels, to compare groundwater level fluctuation in wells, where relative height is defined as:

$$\text{relative height} = \frac{(h_t - h_{initial})}{(h_{\max} - h_{initial})} \quad (7)$$

where h_t = height ASL (m) at time t, $h_{initial}$ = initial groundwater level height ASL (m), and h_{max} = maximum groundwater level height ASL (m).

The rate of groundwater level increase is different for each monitored well, which could be due to a range of factors including aquifer properties, number of RWH structures nearby, and the amount and intensity of rainfall in the local area. The variability highlights local catchment variability, a common feature of semi-arid areas (Pilgrim et al., 1988) and also the local recharge impacts of RWH.

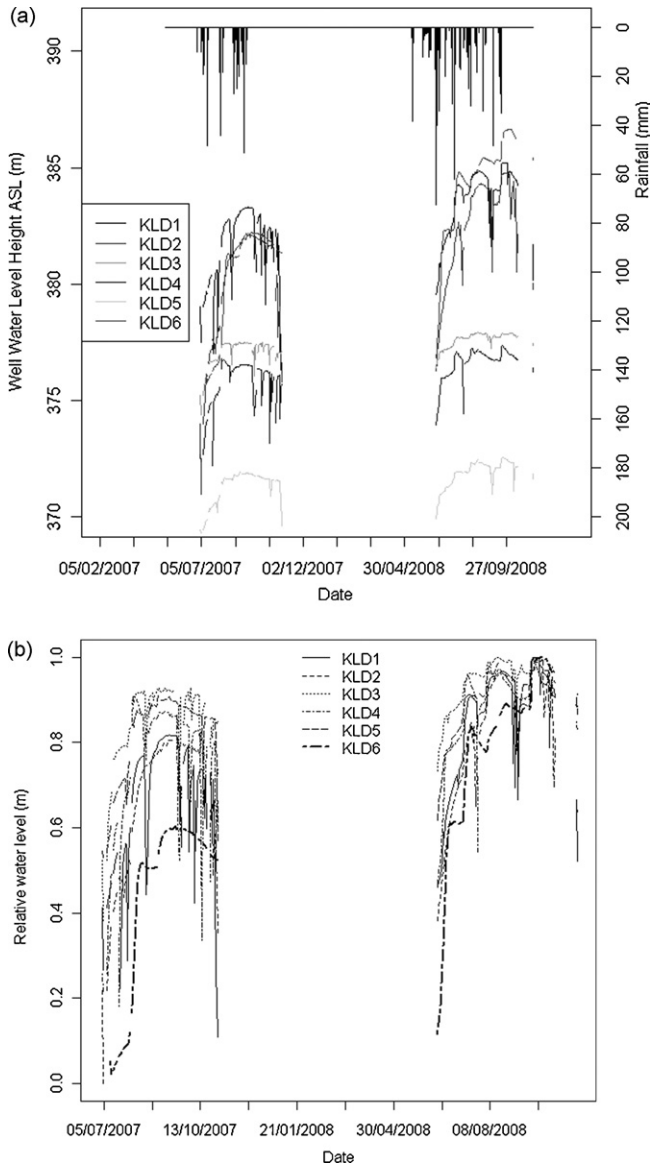


Fig. 8. Response of monitored wells at Kaled: (a) ASL (m) and (b) relative water level height (m).

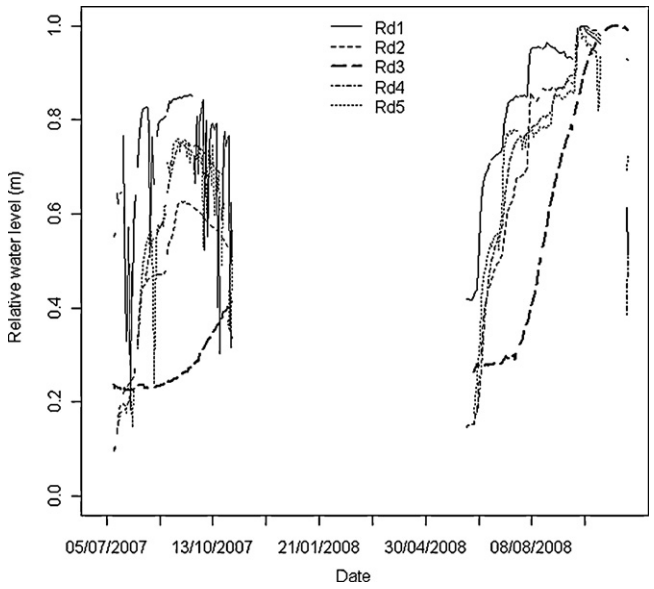


Fig. 9. Roadside wells between Jhiri and Kaled, relative water level height (m).

The wells at Bhaonta saw a greater response in groundwater level in 2008 than in 2007 (Fig. 7a and b), due to less pumping and more rainfall in 2008. The relative responses in 2008 of BK1, BK2, BK4 and BK5 wells show a greater increase than in BK3 and BK6 (Fig. 7b), which could be attributed to their landscape positions. Wells BK3 and BK6 are in a valley that runs parallel to the valley within which the *Bandhs* are found, so recharge may take longer to reach these wells due to differences in hydrogeology. Well BK5 was the shallowest well, close to the surface in a fractured rock area, while BK1 and BK2 were in the agriculture land and thus in alluvial deposits. The response from well BK5 closely mirrored the RWH structures water storage patterns of rise and fall (Fig. 7c), highlighting the direct influence of recharge from RWH on groundwater levels. Wells BK4 and BK 6 are almost 1 km away from Sankara *Bandh* and 600 m from Beruji *Bandh*, but the influence of recharge on groundwater levels in these wells can still be clearly seen. This could be the result of large lateral transmissivity as Bhaonta is in a higher rocky elevation area and there are larger volumes of R_{ep} coming from the *Bandhs*, particularly from Sankara *Bandh*.

At Kaled, wells KLD1–KLD5 are situated along the river below the *Anicut* and display similar characteristics in response (Fig. 8a). Well KLD6 has no pump and is more than 1 km away from the river and as a result the response here was slower and perhaps does not reflect the mounding influence of the *Anicut*, but the overall recharge in the aquifer (Fig. 8b). Well KLD6 also indicates a higher response in 2008 than 2007. This indicates that overall recharge was much higher in 2008 compared to 2007, and that the effects of recharge reach across a wide area.

The roadside wells in Fig. 9 are situated between the Jhiri and Kaled focus areas. These wells were not situated as close to RWH structures as the other wells and might be interpreted as average background values of recharge (Fig. 9). Water levels in well RD3 continued to rise when other wells nearby had already decreased, which most likely represents lateral flow in the shallow alluvial aquifer, rather than natural recharge in the area of the well itself.

4.7. Recharge estimates (R_{gw}) based on the water table fluctuation method

The most unaffected section of the time series, at the start of the monsoon season in both years, was used to estimate R_{gw}. In 2007, due to heavy pumping, the unaffected time series for each well was very short, and for some wells not available. In 2008, however, the unaffected sections of the time series were longer as there was less pumping. Analysis of this section of the time series coincides with the steeper part of the groundwater rises, which means it is likely to result in more accurate estimates of recharge, because it would be less affected by depressions due to pumping.

Due to the lack of information about the extent and properties of the local aquifer it is difficult to extrapolate the values of R_{gw} to a volumetric increase in the aquifer. For illustration purposes the average daily R_{gw} rates (mm/day) for each year were spatially interpolated using the inverse distance weighting method. More detailed soil and geological characteristics would be required to model groundwater recharge accurately, which is not available for this area and beyond the scope of this study. Nevertheless the interpolations indicate that the rates were higher in 2007 than 2008 (Fig. 10). Despite the uncertainty of interpolation, it indicates a general increasing trend in daily R_{gw} moving down the catchment, and to lower elevations. This might suggest flow of groundwater down the catchment dictated by elevation. However, the wells at

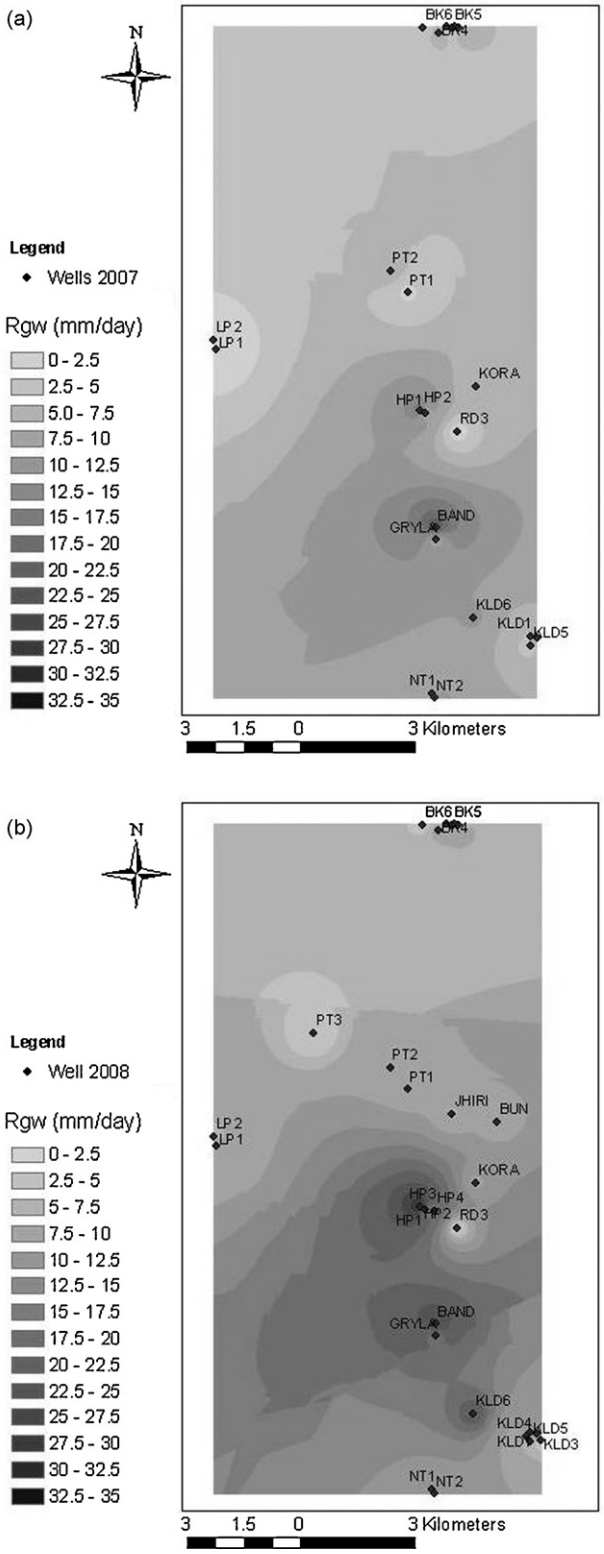


Fig. 10. Interpolations of average recharge in monitored dug wells (R_{gw} mm/day) from WTF in: (a) 2007 and (b) 2008.

Hamirpur and the wells south of the focus area of Hamirpur, have higher recharge rates than at Kaled and Nitata focus areas, which are further downstream (Table 8). This may be due to different aquifer properties in these areas, which are not captured in the WTF method, as a constant S_y was used in the analysis. It might be possible to deduce the lateral movements from the temporal variation in well responses down the catchment. However, due to the spatial and temporal distribution of rainfall such an analysis is not straight forward.

Table 8
Average groundwater recharge R_{gw} (mm/day) in the focus areas of the study in the Arvari River catchment.

Focus areas	Number of monitored wells	Average R_{gw} (mm/day) 2007	Average R_{gw} (mm/day) 2008
Bhaonta	6	4.6	7.2
Lalpura	2	2.5	7.5
Jhiri-Lalpura road	3	4.1	6.9
Jhiri	1	NA	7.3
Hamirpur	4	12.4	23.9
Jhiri-Kaled road	5	9.6	13.1
Kaled	6	7.3	13.0
Nitata	2	10.0	11.5

5. Discussion

5.1. Comparison of recharge estimates

The average daily potential recharge (R_{ep} mm/day) from the monitored RWH structures was greater in 2008 than in 2007 and this increase was also seen in the groundwater level response in the wells (R_{gw} mm/day). However, average daily R_{ep} is much greater than average daily R_{gw} for wells close to the structures (Fig. 11). This difference could be due to either a large amount of lateral flow or due to water stored in the soil profile below the RWH storage area. This is further examined by estimating hydraulic properties of the soils in the RWH storage areas. Based on the field textures and using the soil texture triangle (Saxton et al., 1986), field capacity (FC), available water (AW) and saturated hydraulic conductivity of the soil (K_{sat}) were obtained (Table 9). Using these characteristics, approximately 30% of R_{ep} is held in the soil, and about 13–32% is R_{gw} (Table 9). Approximately 30–50% of the difference therefore could be lateral flow, as groundwater moving down the catchment. Sankara *Bandh* was furthest away from the wells, so a larger lateral flow component is more likely (Table 9). There is possibly less lateral flow from the *Johads* (Jhiri and Lalpura) due to higher storage

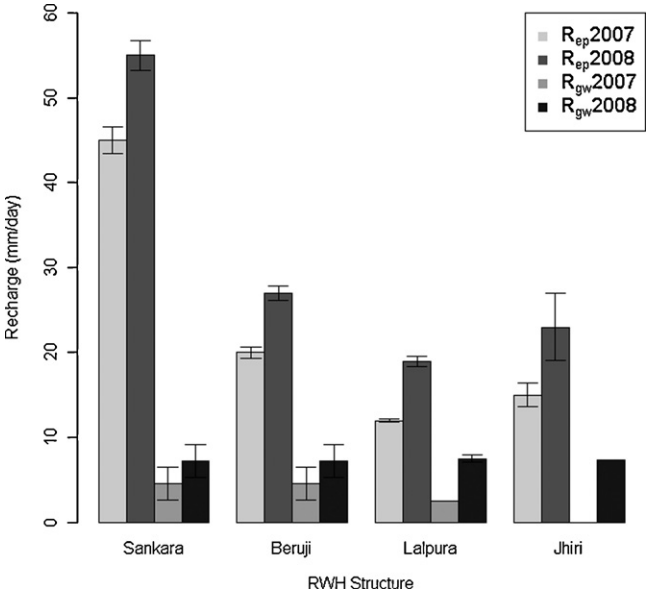


Fig. 11. Comparison of potential recharge estimates from monitored RWH (R_{ep} mm/day) and recharge in monitored dug wells (R_{gw} mm/day) in 2007 and 2008.

Table 9
Hydraulic properties for soil in monitored RWH structures using the soil texture triangle (cm^3 water/cm^3 soil) (Saxton et al., 1986), with an example analysis of the distribution of the recharge R_{ep} (mm) through the soil profile on 1 day in 2008.

Structure	Soil texture	Sand (%)	Clay (%)	FC	AW	K_{sat} (cm/h)	Average R_{ep} (mm)	FC %	R_{ep} as soil water (mm)	R_{ep} – soil water (mm)	R_{gw} (mm)	Lateral flow (mm)	R_{ep} that could be lateral flow (%)
Sankara	Silt loam	20	20	0.29	0.17	0.48	55.6	29	16.1	39.5	7.2	32.3	58
Beruji	Silt loam	20	20	0.29	0.17	0.48	27.8	29	8.0	19.7	7.2	12.5	45
Jhiri	Sandy clay loam	50	30	0.28	0.11	0.29	19.5	28	5.5	14.0	7.3	6.7	35
Lalpura	Sandy clay	50	40	0.32	0.10	0.15	23.5	32	7.5	16.0	7.5	8.5	36

in the soil and lower recharge values. Larger local mounds, such as possibly under Sankara *Bandh*, would also increase hydraulic gradients and thus increase lateral movement of water.

The high lateral flow also suggests that rise in wells further away from RWH structures is more likely to be recharge from RWH structures than natural recharge. Natural recharge estimates from the regression developed by Rangarajan and Athavale (2000) for alluvial areas gives approximately 57 mm/year of natural recharge in 2007 and 104 mm/year of natural recharge in 2008. These estimates are smaller than estimated R_{gw}, which ranges from approximately 200 to 300 mm in the monsoon season.

5.2. Potential recharge (R_{ep}) from RWH in the catchment

The total approximate maximum storage area of RWH structures is 1.44% of the Arvari River catchment area. The RWH storage surface area was considered to be the same for 2007 and 2008 for the calculation presented in Table 10. The rainfall at Hamirpur was used to calculate the percentage of rainfall that becomes recharge from the similar sized *Anicuts*/*Bandhs* and *Johads*/*Talab* in the catchment (Table 10) at the catchment scale. In 2007 around 6.6% of rainfall became R_{ep}, and in 2008 about 7%. These values are actually quite similar to recharge efficiency estimates from the monitored RWH structures (Table 7). The fraction of rainfall that becomes recharge is also very similar between a dry and a wet year, which reflects the recharge efficiency of the structures.

Estimates of R_{ep} from RWH structures generally agree with Sharda et al. (2006) and Badiger et al. (2002) that less than 10% of rainfall becomes R_{ep}. Interestingly while R_{ep} amounts were higher in 2008, recharge efficiencies were similar in both years. Efficiency is a function of infiltration and structure shape and size rather than rainfall amount. These factors also influenced the amount of rainfall that becomes R_{ep}.

5.3. Behaviour of recharge between different rainfall years

In a data sparse environment such as the study area, first estimates of catchment hydrological behaviour, particularly influenced by RWH, are an important step towards understanding the existing processes in the catchment. The high temporal and spatial variability of rainfall is typical of monsoonal rainfall in Rajasthan where rainfall is characterised by short intensity and highly localised (Sharma and Murthy, 1995).

The results of this study are limited by the length of the time series available, but it compares a comparatively wet and dry year indicating the variation in possible R_{ep} from RWH structures. The impact of higher annual rainfall on recharge can be clearly seen in 2008, with higher rainfall resulting in more overall potential recharge. Despite this variation in rainfall, the fraction of rainfall that becomes recharge was very similar between the years reflecting the similar recharge efficiency of the structures. A difference in R_{ep} was found between *Bandhs* and *Johads*, with an average R_{ep} of 37.5 mm/day for *Bandhs*, and 17.8 mm/day for *Johads*. It must be noted that the potential recharge rates from RWH structures may change over time. They could decrease due to siltation, which occurs when particles brought by runoff are deposited in the structures and build over time to form a layer with lower hydraulic conductivity. In this catchment, each village has formed a *Gram Sabha* (village council), which undertakes regular maintenance of the RWH structures, including desilting. Nevertheless the impact of siltation on the recharge capacity of RWH structures, and the frequency of desilting required to maintain effective recharge is a needed area of future research.

In both years of the analysis, clustered rainfall events and larger storms were needed to generate more stream flow and stream flow was intermittent. The estimated R_{ep} values from the *Anicuts* were high compared with *Bandhs* and *Johads*. This confirms the local knowledge that the *Anicuts* provide a great amount of recharge. It also suggests substantial transmission losses from the river (Lange, 2005).

By taking the most unaffected sections of the time series, and assuming a constant specific yield, estimates of R_{gw} ranged from 7.2 to 11.3 mm/day. The recharge values derived from the well data were much lower than the estimates of R_{ep} from the RWH structures, suggesting that the aquifer has a large lateral transmissivity. About 30% of R_{ep} is stored in the soil while at least another 30% moves laterally. As a result R_{ep} from the structures would initially cause local mounding, which then dissipates across the aquifer. If lateral flow is significant, then the downstream impacts of reducing surface runoff upstream due to RWH could result in a beneficial increase in groundwater levels downstream as a result of lateral flow. Quantification of such flow would require greater knowledge of aquifer properties, which is limited in this catchment. Nevertheless the use of the water balance approach in structure storages, combined with the water table fluctuation method in dug wells nearby, provides first estimates of local-scale impact of RWH.

Table 10
Approximate area of the catchment for different RWH structure types in the Arvari River catchment and approximate percentage of annual rainfall that becomes recharge R_{ep} in 2007 and 2008.

RWH structure type	% area of catchment	% annual rainfall 2007	% annual rainfall 2008
Anicut/*Bandh*	0.82	9.5	9.1
Johad/*Talab*	0.61	2.4	4.0
Total	1.44	6.6	7.1

6. Conclusions

Increasingly, rainwater harvesting is being implemented in India to alleviate declining groundwater stores, but there is a need for better understanding of the impacts of many small dams within a catchment. Unfortunately RWH is often found in remote regions, where data is limited, restricting hydrological studies of RWH impact. In this catchment, there is no climate station and there is a dearth of information about local-scale soil and geological characteristics. Even so, the field data from this study highlights the temporal and spatial variability of the hydrological processes associated with RWH in a semi-arid catchment. Structure types exhibit similar recharge characteristics. In general only about 7% of rain-

fall is transformed to recharge and recharge efficiencies are similar across years. Rainwater harvesting structures clearly have a large impact on the amount of recharge occurring in a local area and RWH function is dependent on rainfall, with larger R_{ep} volumes in the higher rainfall year. However, the size of the storage area in the structure influences the maximum daily depth of recharge, which reaches a maximum limit. To model the impact of RWH will require data on the number, physical specifications, and types of structures that exist in the catchment, rainfall characteristics, stream flow and aquifer properties.

Acknowledgements

The authors would like to thank the NGO, Tarun Bharat Sangh, for support, and Murari Lal Gujjarji, who provided valuable field assistance. The comments of two anonymous reviewers are also appreciated.

References

Badiger, S., Sakthivadivel, R., Aloysius, N., Sally, D., 2002. Preliminary assessment of a traditional approach to rainwater harvesting and artificial recharging of groundwater in Alwar District, Rajasthan. In: Annual Partners' Meet 2002, IWMI-TATA Water Policy Research Program, pp. 1–18.

Barah, B.C., 1996. Traditional Water Harvesting Systems: An Ecological Economic Survey. New Age International Limited, New Delhi.

Batchelor, C., Rama Mohan Rao, M.S., Manohar Rao, S., 2003. Watershed development: a solution to water shortages in semi-arid India or part of the problem? Land Use and Water Resources Research 3, 1–10.

Batchelor, C., Singh, A., Mohan Rao, R., Butterworth, J., 2002. Mitigating the potential unintended impacts of water harvesting. In: IWRA International Regional Symposium "Water For Human Survival", Hotel Taj Palace, New Delhi, India.

Beernaerts, I., 2006. An evaluation Framework for Groundwater Recharge Enhancement Systems for Improved Irrigated Agriculture. Division of Information Technology, Engineering and the Environment, University of South Australia, Adelaide, p. 86.

Bhuiyan, C., Singh, R.P., Kogan, F.N., 2006. Monitoring drought dynamics in the Aravalli region (India) using different indices based on ground and remote sensing data. International Journal of Applied Earth Observation and Geoinformation 8, 289–302.

Calder, I., Gosain, A., Rama Mohan Rao, M.S., Batchelor, C., Garratt, J., Bishop, E., 2006. Watershed development in India. 2. New approaches from managing externalities and meeting sustainability requirements. Environment, Development and Sustainability.

COMMAN, 2005a. Managing groundwater resources in Rural India: the community and beyond. British Geological Survey Commissioned Report, CR/05/35N.

COMMAN, 2005b. Community management of groundwater resources in Rural India: background papers on the causes, symptoms and mitigation of groundwater overdraft in India. British Geological Survey Commissioned Report CR/05/36N.

Crosbie, R.S., Binning, P., Kalma, J.D., 2005. A time series approach to inferring groundwater recharge using the water table fluctuation method. Water Resources Research 41.

Gale, I., Neumann, I., Calow, R., Moench, M., 2002. The effectiveness of artificial recharge of groundwater: a review. Commissioned Report CR/02/108N, British Geological Survey, Nottingham, pp. 1–50.

Gontia, N.K., Sikarwar, R.S., 2005. Rainwater harvesting and groundwater recharge in Saurashtra region of Gujarat—a success story. Indian Journal of Soil Conservation 33, 256–258.

Gupta, S.K., Deshpande, R.D., 2004. Water for India in 2050: first-order assessment of available options. Current Science 86, 1216–1224.

Healy, R., Cook, P., 2002. Using groundwater levels to estimate recharge. Hydrogeology Journal 10, 91–109.

India Water Portal, 2009. Meteorological Data—CRU TS 2.1. In: Mitchell, T.D., Jone, P.D. (Eds.), Tyndall Center for Climate Change Research. University of East Anglia.

Johnson, A.I., 1967. Specific yield—compilation of specific yields for various materials. U.S. Geological Survey Water Supply Paper, p. 74.

Keller, A., Sakthivadivel, R., Seckler, D., 2000. Water scarcity and the role of storage in development. Research Report Number 39, International Water Management Institute, Colombo, Sri Lanka.

Kemper, K.E., 2007. Instruments and institutions for groundwater management. In: Giordano, M., Villholth, K. (Eds.), The Agricultural Groundwater Revolution. Opportunities and Threats to Development. CAB International Publishing, Colombo, Sri Lanka.

Kumar, M.D., Patel, A., Ravindranath, R., Singh, O.P., 2008. Chasing a mirage: water harvesting and artificial recharge in naturally water-scarce region. Economic and Political Weekly 43, 61–70.

Kumar, M.D., Ghosh, S., Patel, A., Singh, O.P., Ravindranath, R., 2006. Rainwater harvesting in India: some critical issues for basin planning and research. Land Use and Water Resources Research 6, 1–17.

Lange, J., 2005. Dynamics of transmission losses in a large arid stream channel. Journal of Hydrology 306, 112–126.

Marechal, J.C., Dewandel, B., Ahmend, S., Galeazzi, L., Zaidi, F.K., 2006. Combined estimation of specific yield and natural recharge in a semi-arid groundwater basin with irrigated agriculture. Journal of Hydrology 329, 281–293.

Martin-Rosales, W., Gisbert, J., Pulido-Bosch, A., Vallejos, A., Fernandez-Cortes, A., 2007. Estimating groundwater recharge induced by engineering systems in a semiarid area (South Eastern Spain). Environmental Geology 52, 985–995.

Muralidharan, D., Andrade, R., 2007. Evaluation of check-dam recharge through water-table response in ponding area. Current Science 92, 1349–1351.

Narain, P., Khan, M.A., Singh, G., 2005. Potential for water conservation and harvesting against drought in Rajasthan, India. Working Paper 104 (Drought Series: Paper 7), International Water Management Institute, Colombo, p. 25.

Neumann, I., MacDonald, D., Gale, I., 2004. Numerical approaches for approximating technical effectiveness of artificial recharge structures. Commissioned Report CR/04/265N, British Geological Society, Keyworth, Nottingham, 46 pp.

Pandey, D.N., Chaubey, A.C., Gupta, A.K., Vardhan, H., 2005. Mine spoil restoration: a strategy combining rainwater harvesting and adaptation to random recurrence of droughts in Rajasthan. International Forestry Review 7, 241–249.

Pilgrim, D.H., Chapman, T.G., Doran, D.G., 1988. Problems of rainfall-runoff modelling in arid and semiaird regions. Hydrological Sciences Journal 33, 379–401.

Ramesh, R., Yadava, M.G., 2005. Climate and water resources of India. Current Science 89, 818–824.

Rangarajan, R., Athavale, R., 2000. Annual replenishable ground water potential of India—an estimate based on injected tritium studies. Journal of Hydrology 234, 38–53.

Rodell, M., Velicogna, I., Famiglietti, J.S., 2009. Satellite-based estimates of groundwater depletion in India. Nature 460.

Sakthivadivel, R., 2007. The groundwater recharge movement in India. In: Giordano, M., Villholth, K. (Eds.), The Agricultural Groundwater Revolution. Opportunities and Threats to Development. CAB International Publishing, Colombo, Sri Lanka.

Sakthivadivel, R., 2008. Decentralized artifical recharge movements in India: potential and issues. In: Amarasinghe, U.A., Sharma, B.R. (Eds.), Strategic Analyses of the National River Linking Project (NRLP) of India Series 2. International Water Management Institute, pp. 315–325.

Saxton, K.E., Rawls, W.J., Romberger, J.S., Papendick, R.I., 1986. Estimating generalized soil–water characteristics from texture. Soil Science Society of America Journal 50, 1031–1036.

Scanlon, B., Healy, R., Cook, P., 2002. Choosing appropriate techniques for quantifying groundwater recharge. Hydrogeology Journal 10, 18–39.

Shah, T., 2007. The groundwater economy of South Asia: an assessment of size, significance and socio-ecological impacts. In: Giordano, M., Villholth, K. (Eds.), The Agricultural Groundwater Revolution. Opportunities and Threats to Development. CAB International Publishing, Colombo, Sri Lanka.

Shah, T., Singh, O.P., Mukherji, A., 2006. Some aspects of South Asia's groundwater irrigation economy: analyses from a survey in India, Pakistan, Nepal Terai and Bangladesh. Hydrogeology Journal 14, 286–309.

Shah, T., Burke, J., Villholth, K., 2007. Groundwater: a global assessment of scale and significance. In: Molden, D. (Ed.), Water for Food, Water for Life: A Comprehansive Assessment of Water Management in Agirculture. Earthscan.

Sharda, V.N., Kurothe, R.S., Sena, D.R., Pande, V.C., Tiwari, S.P., 2006. Estimation of groundwater recharge from water storage structures in a semi-arid climate of India. Journal of Hydrology 329, 224–243.

Sharma, K.D., Murthy, J.S.R., 1995. Hydrologic routing of flow in arid ephemeral channels. Journal of Hydraulic Engineering, 466–471.

Sophocleous, M., 1991. Combining the soil water balance and water-level fluctuation methods to estimate natural groundwater recharge: practical aspects. Journal of Hydrology 124, 229–241.

Villholth, K., 2006. Groundwater assessment and management: implications and opportunities of globalization. Hydrogeology Journal 14, 330–339.

Wendland, E., Barreto, C., Gomes, L.H., 2007. Water balance in the Guarani Aquifer outcrop zone based on hydrogeologic monitoring. Journal of Hydrology 342, 261–269.